Aligning the Greedy Operating Point of a Generative Chest X-ray VLM with its Discriminative Optimum: Stage-3 Training via Diverse Multiturn Composition and Asymmetric Token Weighting

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Abstract

A jointly trained chest X-ray vision–language model (Qwen3.5-2B with an ML-Decoder classification head) exhibited a large and systematic gap between its *discriminative* ability and its *generative* behaviour: teacher-forced presence log-probabilities achieved a nodule ROC-AUC of 0.830 — statistically indistinguishable from the dedicated 132-tag classification head (0.836) — yet free-running greedy generation asserted `presence: true` for only 13.2% of ground-truth nodule cases. The model *knew*, but would not *say*. We describe a third supervised fine-tuning stage (S3) whose explicit objective was to move the greedy operating point of the generated classification JSON onto the Youden-optimal point of the model's own underlying score — the point the ML-Decoder head reaches when thresholded at sensitivity = specificity. S3 combines (i) a 403,820-conversation multiturn dataset composed from 19 task-chain templates over previously trained capabilities, (ii) a frozen vision tower and frozen classification head to preserve the validated discriminative substrate, and (iii) an asymmetric 6:1 cross-entropy weight on the `true`/`false` presence tokens, implemented at zero cost inside the fused Liger kernel. On 17,880 production studies, free-running nodule-family sensitivity rose from 0.292 to 0.588 at specificity 0.931; the Youden index rose +83% (0.284→0.519), reaching parity with the head's equal-error operating point on several findings (calcification J=0.534 vs. head 0.534; atelectasis 0.437 vs. 0.448) while preserving the head's ROC-AUC exactly (frozen weights, bit-identical). Macro Youden across 194 findings improved +48%. We argue that single-point Youden analysis — not AUC — is the correct lens for evaluating generative classifiers, and derive why a 6:1 token weight approximates the Youden threshold at the prevalence regime of this dataset.

1 Introduction

Structured-report generation and multi-label classification are usually treated as separate problems. Our production CXR system trains them *jointly*: a 2.7B-parameter Qwen3.5-2B VLM generates a detailed classification JSON (per-finding `presence`/`size`/`location`/`characteristics` over a ∼112-finding ontology), Answer-Set-Programming (ASP) reasoning traces, knowledge graphs, and full radiology reports, while a 12.8M-parameter ML-Decoder head attached to the frozen-at-inference vision tower produces 132 calibrated tag scores from the same visual features.

Evaluated on 136,639 production studies with report-derived silver labels, the two faces of the same model told contradictory stories:

- **Discriminatively**, the model was strong. The ML-Decoder head reached a macro ROC-AUC of **0.885** (nodules 0.836, granuloma 0.930, pneumothorax 0.970). Teacher-forced presence log-probabilities of the LLM itself — reading p(`true`) at each finding's value slot — yielded a nodule AUC of **0.830**, tied with the dedicated head, and within 0.01–0.06 of it on most parenchymal findings.
- **Generatively**, the model was almost mute. Under greedy decoding the free-running classification JSON asserted nodules with sensitivity **0.132** (family-level 0.292), pneumonia 0.185, pulmonary edema 0.152, rib fractures 0.152 — at specificities of 0.99+. Macro sensitivity over the shared vocabulary was 0.164.

The information was demonstrably present in the network's token distribution; greedy argmax simply never crossed the lexical decision boundary. This paper documents the third training stage (S3) built to close that gap, its measured effect, and the operating-point theory that motivated both its design and its evaluation.

2 The Operating-Point Problem

2.1 Greedy generation is a fixed, accidental threshold

Let $m(x) = \log p(\texttt{true} \mid x) - \log p(\texttt{false} \mid x)$ be the logit margin at the presence-value slot for a finding in study x. Teacher-forced AUC evaluates the *ranking* induced by m over studies — it integrates over all thresholds and is therefore threshold-free. Greedy generation, by contrast, emits `true` iff $m(x) > 0$: it is a *single* operating point, fixed at margin zero by the token prior the model absorbed during training.

Two properties of the pre-S3 training data pinned that threshold in a pathological place. First, the classification-JSON targets are overwhelmingly negative (∼6 true among ∼112 findings per study), so the unconditional prior of the `true` token at any given slot is ∼5%: the model must overcome a ∼3-nat prior deficit before *asserting* a finding. Second, symmetric cross-entropy makes every token error equally costly, so the loss-minimising behaviour at ambiguous slots is to emit the majority token. The result is exactly what we measured: a classifier with AUC 0.83 operating at the extreme-specificity corner of its own ROC curve (sens 0.13 / spec 0.996 for nodules).

2.2 Why Youden's J, not AUC, is the right lens

For a *generated* binary decision there is no score to sweep: ROC-AUC is undefined. The two familiar summaries of a single operating point are balanced accuracy $\mathrm{bAcc} = (\mathrm{Se} + \mathrm{Sp})/2$ and Youden's index $J = \mathrm{Se} + \mathrm{Sp} - 1 = 2\,\mathrm{bAcc} - 1$. J has three properties that make it the correct target here:

1. **It is chance-corrected and threshold-honest.** Any trivial threshold slide along the chance diagonal leaves J=0; sensitivity bought one-for-one with specificity does not move it. A rise in J certifies that the operating point moved *toward the ROC curve's shoulder*, i.e. that usable separation was gained.
2. **Its maximum over thresholds is a property of the score, not the threshold:** $J^* = \max_\tau \big(\mathrm{Se}(\tau) + \mathrm{Sp}(\tau) - 1\big)$ is the Youden-optimal point of the underlying ROC. For the ML-Decoder head we can compute J^* directly (its equal-error-rate point closely approximates it); for the LLM's teacher-forced log-probability score, the same construction applies. **These two J^* values are the ceiling that any decoding strategy can reach without changing the model's knowledge.**
3. **It composes into the deployment question.** A generative report is only usable if what it *asserts* sits at a clinically sensible point; J measures precisely the quality of the assertion rule.

This framing yields the quantitative goal of S3, stated before training began: *move the greedy operating point of the generated JSON to $J_{\text{greedy}} \approx J^*$, where J^* is the maximum Youden index of the model's own discriminative scores — equivalently, of the ML-Decoder head (AUC 0.885) and the LLM's log-prob AUC (0.83 nodules), which are empirically the same signal.* Since the head thresholded at its EER point realises J within 1–2% of its J^*, the head's EER row in our tables serves as the concrete, measurable target.

3 Stage-3 Training Strategy

3.1 Substrate: what S1 and S2 built

S1 (13.3M prompts over 4.8M conversations) taught nine task families — sectioned and structured report generation (9 sub-tasks × 1.06M), single- and multi-image bounding-box grounding (2.3M), ASP reasoning (0.87M), cross-image report comparison (0.49M), and focused COPD/nodule/device analyses. S2 added the full-ontology classification JSON with *explicit negatives* (crucially: earlier data contained zero `presence: false` fields, making the presence token a syntactic constant) and jointly trained the ML-Decoder head with an asymmetric-loss (ASL) objective on 132 tags, reaching head macro AUC 0.885 with no loss of generative ability.

3.2 Data: 19-template multiturn composition (403,820 conversations)

S3's dataset was *composed*, not collected: every conversation chains existing capabilities on the *same image*, joined across the S1/S2 sources by image key. Eight single-image chains start with the classification JSON and append grounding, ASP, knowledge-graph, ontology-report, report-section, or follow-up turns (templates A–H); six more end in the *structured JSON report* with two non-image-derivable sections (*Identification and*

Administrative Data, *Comparison with Previous Images*) removed from every target (N–S); two novel *region-identify* templates invert grounding (given 1–3 box coordinates, name the abnormality; T,U); and three multi-image templates cover per-image classification followed by report comparison, plus bbox-comparison and nodule-bbox replay (I,K,M). Images are allocated *disjointly* across templates; 100 rows of every template form a stratified validation set (1,900 rows). Two data repairs were applied during composition: sub-100 px^2 degenerate boxes were dropped, and the `atelactasis` spelling variant (234k instances) was normalised in box labels.

The rationale: chained turns teach the model that the classification JSON, the ASP trace, the graph, and the report are *views of one committed read* — the report turn is trained to condition on an *actually generated* classification rather than on nothing — while template diversity prevents any single chain from becoming a rote sequence.

3.3 Freezing the discriminative substrate

The vision tower (306M parameters) and the ML-Decoder head (12.8M) were **frozen**; only the LLM (2.395B) and the vision–language projector (20M) trained (88.3% of parameters). The classification-head loss was disabled (ASL_WEIGHT=0), as a frozen head admits no gradient path. Freezing serves the paper's central logic: *the discriminative optimum J^* must remain a fixed target while the generative policy moves toward it.* Any drift of the visual features would change both the head's AUC and the LLM's log-prob AUC, making before/after Youden comparisons unattributable. Operationally, the head's 23 tensors are re-loaded from the S2 checkpoint at attach time (they would otherwise be silently re-initialised) and are carried bit-identically into every saved checkpoint, so downstream head-based inference is unchanged. A corollary requirement: geometric image augmentation (elastic/grid distortion, 10–15% crop-zoom) would silently corrupt box-coordinate supervision, so augmentation is suppressed for any conversation containing `bbox_2d` (∼53% of rows) and retained photometrically-and-geometrically for the rest.

3.4 Asymmetric token weighting: engineering the threshold shift

The decisive intervention is a class-conditional weight on the AR cross-entropy, implemented through the fused Liger kernel's `ce_weight` vector (gathered by target-token id; zero memory and zero throughput cost, in contrast to any `loss_scale` path that would materialise a ∼10 GB logits tensor):

$$w(\texttt{true}) = 6, \qquad w(\texttt{false}) = 1, \qquad w(\text{JSON scaffolding}) = 0.5, \qquad w(\text{other}) = 1.$$

Consider the token-level decision in isolation. Weighted cross-entropy at a presence slot is minimised by asserting `true` whenever

$$w_t\, p^*(\texttt{true} \mid x) > w_f\, p^*(\texttt{false} \mid x) \iff p^*(\texttt{true} \mid x) > \frac{1}{1 + w_t/w_f} = \tfrac{1}{7} \approx 0.143,$$

i.e. training pressure moves the greedy boundary from p^*=0.5 to p^*≈0.143 — a ≈1.8-nat shift of the decision margin. For a well-ranked score under class prevalence π, the Youden-optimal threshold sits near the point where the weighted densities cross; with the 5–10% prevalence typical of our mid-frequency findings and the empirical score distributions of the S2 model, the EER/Youden point corresponds to accepting positives down to roughly $p^* \sim 0.1$–0.2 — which is precisely the region the 6:1 weight targets. The weight is thus not a heuristic "recall boost" but a calculated re-anchoring of the generative threshold onto J^*. (S2 had deliberately trained these tokens at 1:1; its greedy point sat at margin 0, explaining the whole pathology of Section 2.1.)

3.5 Optimisation

One epoch over 401,920 conversations; effective batch 1,024 (8×H200-class GPUs, per-device batch 8, gradient accumulation 16); LLM LR 4×10^{-6}, projector 5×10^{-5}; bf16, ZeRO-2, Liger fused CE, gradient checkpointing, no packing (image→row correspondence must survive for the multiturn/box supervision); 393 optimiser steps, 11.7 h wall-clock. Eval loss on the stratified set fell 0.128→0.124 and plateaued from step ∼250; eval token accuracy was flat at 95.9% throughout — foreshadowing that the improvement lives in a handful of decision tokens, invisible to token-level accuracy but decisive for J.

4 Results

4.1 Headline: the greedy point reached the head's balanced point

All S2/S3 numbers below are free-running greedy generation, no head guidance, on the same 17,880 production studies (interim cut of the full 136k evaluation; the ordering of studies is arbitrary so this is an unbiased sample). Head numbers are its EER (sens=spec) operating point on the full 126,856-study set, i.e. its realised $J \approx J^*$.

Table 1: Nodule family (nodule ∨ mass ∨ granuloma) and macro summary. J = Youden index; bAcc = balanced accuracy. Head at its sens=spec point.

	Sens	Spec	J	bAcc	ΔJ vs S2
S2 greedy	0.292	0.992	0.284	0.642	—
S3 greedy	**0.588**	**0.931**	**0.519**	**0.759**	**+83%**
Head @ EER (target)	0.775	0.775	0.550	0.775	—
Macro, 194 tags: S2	0.164	0.992	0.155	0.578	—
Macro, 194 tags: **S3**	**0.257**	**0.973**	**0.229**	**0.614**	**+48%**

The nodule-family greedy J moved from 52% to **94% of the head's realised** J. On individual findings the target was fully reached or exceeded — calcification J 0.534 vs. head 0.534 (exact tie), atelectasis 0.437 vs. 0.448, consolidation 0.431 vs. 0.456, pleural effusion 0.703 vs. 0.718 — while others retain headroom (nodules 0.426 vs. 0.518; pneumothorax 0.753 vs. 0.829; pulmonary edema 0.477 vs. 0.587). Devices, already at ceiling in S2, were untouched (ETT J 0.897→0.906 vs. head 0.952), confirming that the intervention shifted under-called findings without destabilising solved ones.

Table 2: Per-finding comparison, findings with ≥450 GT positives in the interim set. Each cell: S2 → **S3**, head EER point in ⟨·⟩. Percentage = relative J gain S2→S3.

Finding	n	Sens	Spec	J	bAcc	ΔJ
atelectasis	5776	.416→**.806** ⟨.723⟩	.900→**.631** ⟨.725⟩	.316→**.437** ⟨.448⟩	.658→**.718** ⟨.724⟩	+38%
pleural effusion	4771	.714→**.890** ⟨.858⟩	.939→**.814** ⟨.859⟩	.653→**.703** ⟨.718⟩	.827→**.852** ⟨.859⟩	+8%
calcification	3071	.290→**.801** ⟨.767⟩	.970→**.733** ⟨.767⟩	.260→**.534** ⟨.534⟩	.630→**.767** ⟨.767⟩	+105%
cardiomegaly	3069	.520→**.916** ⟨.804⟩	.937→**.631** ⟨.802⟩	.457→**.547** ⟨.606⟩	.729→**.773** ⟨.803⟩	+20%
pneumonia	2563	.185→**.721** ⟨.699⟩	.976→**.647** ⟨.699⟩	.161→**.368** ⟨.399⟩	.580→**.684** ⟨.699⟩	+129%
fibrosis	1724	.206→**.584** ⟨.765⟩	.976→**.853** ⟨.764⟩	.182→**.437** ⟨.529⟩	.591→**.719** ⟨.764⟩	+140%
pulm. vasc. congestion	1685	.084→**.569** ⟨.757⟩	.992→**.862** ⟨.758⟩	.075→**.431** ⟨.515⟩	.538→**.716** ⟨.757⟩	+474%
pulmonary edema	1508	.152→**.603** ⟨.794⟩	.987→**.874** ⟨.792⟩	.139→**.477** ⟨.587⟩	.569→**.739** ⟨.793⟩	+244%
ILD pattern	1465	.156→**.414** ⟨.734⟩	.993→**.912** ⟨.734⟩	.148→**.326** ⟨.468⟩	.574→**.663** ⟨.734⟩	+119%
endotracheal tube	1319	.918→**.928** ⟨.976⟩	.979→**.978** ⟨.976⟩	.897→**.906** ⟨.952⟩	.949→**.953** ⟨.976⟩	+1%
consolidation	1242	.280→**.841** ⟨.727⟩	.936→**.589** ⟨.729⟩	.216→**.431** ⟨.456⟩	.608→**.715** ⟨.728⟩	+99%
pneumothorax	977	.529→**.778** ⟨.914⟩	.995→**.975** ⟨.914⟩	.524→**.753** ⟨.829⟩	.762→**.876** ⟨.914⟩	+44%
nodules	917	.132→**.473** ⟨.759⟩	.996→**.953** ⟨.759⟩	.128→**.426** ⟨.518⟩	.564→**.713** ⟨.759⟩	+234%
pleural thickening	915	.108→**.531** ⟨.784⟩	.996→**.935** ⟨.784⟩	.104→**.466** ⟨.569⟩	.552→**.733** ⟨.784⟩	+349%
diaphragm elevation	880	.411→**.741** ⟨.862⟩	.993→**.926** ⟨.862⟩	.404→**.667** ⟨.723⟩	.702→**.834** ⟨.862⟩	+65%
low lung volumes†	869	.284→**.183** ⟨.755⟩	.963→**.977** ⟨.753⟩	.247→**.160** ⟨.508⟩	.624→**.580** ⟨.754⟩	−35%
hyperinflation	761	.175→**.667** ⟨.797⟩	.993→**.885** ⟨.798⟩	.168→**.552** ⟨.595⟩	.584→**.776** ⟨.797⟩	+229%
COPD	729	.228→**.689** ⟨.792⟩	.990→**.868** ⟨.792⟩	.218→**.557** ⟨.584⟩	.609→**.778** ⟨.792⟩	+156%
scoliosis	680	.397→**.716** ⟨.854⟩	.992→**.938** ⟨.855⟩	.389→**.654** ⟨.708⟩	.694→**.827** ⟨.854⟩	+68%
rib fractures	677	.152→**.555** ⟨.784⟩	.998→**.960** ⟨.784⟩	.150→**.515** ⟨.567⟩	.575→**.757** ⟨.784⟩	+244%
granuloma	677	.377→**.659** ⟨.850⟩	.994→**.968** ⟨.849⟩	.371→**.627** ⟨.699⟩	.685→**.814** ⟨.849⟩	+69%
healed rib fracture	457	.195→**.665** ⟨.856⟩	.998→**.971** ⟨.856⟩	.193→**.636** ⟨.712⟩	.597→**.818** ⟨.856⟩	+230%

† the single substantive regression; see §5.

4.2 Why the gain is so large

The magnitude of the improvement — triple-digit relative J gains on ten findings — follows directly from the mechanism. S2's greedy point sat at the margin-0 corner of a *good* ROC curve: the steepest part of the curve, where a small threshold relaxation buys sensitivity almost for free. The 1.8-nat boundary shift engineered by the 6:1 weight traverses exactly that steep region. Findings whose S2 sensitivity was near zero but whose log-prob AUC was high (pulmonary vascular congestion: sens 0.084, head AUC 0.833) gained the most (+474% J), because the discriminative signal was fully formed and purely *gated* by the token prior. Findings already near their optimum (effusion, devices) gained little, exactly as the theory predicts. Three

secondary contributors: (i) chained-conversation training forces the assertion set to be *used* downstream (ASP, report), penalising silent omissions; (ii) explicit-negative targets from S2 remain in the mixture, so the model does not degenerate into over-assertion — specificity stayed at 0.93–0.98 on most findings; (iii) the frozen substrate means all improvement is attributable to the policy/verbaliser, not to representation drift.

4.3 Verification of substrate integrity and format compliance

The frozen head is bit-identical (restored 23/23 tensors; no optimiser group), so head AUC remains 0.885 by construction — verified by re-scoring. Structured-report chains generated by the S3 checkpoint are 100/100 valid JSON with the two excised sections absent in 100% of cases; classification→report finding agreement is 96% (cls→report) and 92% across the longer cls→ASP→report chain. The novel region-identify capability answers with correctly echoed coordinates in 100% of sampled query boxes.

5 Discussion and Limitations

Distance to the ceiling. At family level the greedy J reached 94% of the head's realised J; per-finding, mid-prevalence findings still leave 0.05–0.15 of J on the table (nodules 0.43 vs. 0.52). Uniform 6:1 was calibrated for the prevalence mode, not per finding; the residual gap concentrates where prevalence or score sharpness deviates from that mode. A per-finding weight schedule — or, at inference, per-finding log-prob thresholds read off the (unchanged) teacher-forced margins — would close most of it. **Where the trade bit.** High-prevalence findings over-shot: cardiomegaly (spec 0.63) and consolidation (0.59) now over-assert; their Youden still improved, but a deployment-facing configuration may prefer 4:1 for these tags. **Regressions.** `low_lung_volumes` lost sensitivity (-35% J), plausibly a vocabulary-competition effect with sibling phrasing; and `opacity` remains at $J{=}0$ for both stages because the model prefers specific opacity terms over the umbrella token — a lexical-mapping artefact of the silver labels, not a perception failure. **Evaluation caveats.** Ground truth is an LLM extraction of radiologist reports (silver, report-derived, with known prior-study contamination); S2/S3 numbers are an interim 17,880-study cut of the running 136k evaluation (the head row uses the completed full-set scoring); and the head's EER thresholds were fit on the evaluation distribution, flattering the target slightly. The full-set refresh will finalise all S2/S3 columns.

6 Conclusion

Stage-3 shows that for a jointly trained generative-and-discriminative VLM, the gap between "what the model knows" (AUC of its scores) and "what the model says" (greedy assertions) is a *controllable training variable*, not an inherent cost of generation. Composing multiturn chains over frozen discriminative substrate and re-anchoring the presence-token decision boundary with an asymmetric 6:1 CE weight moved the free-running operating point from the mute corner of the ROC curve ($J{=}0.28$, nodule family) to near the Youden optimum of the model's own scores ($J{=}0.52$ vs. ceiling ≈ 0.55), at a training cost of 11.7 GPU-node-hours and with the classification head — and therefore every head-based downstream consumer — untouched. Youden's index, being exactly the quantity that separates genuine gained separation from threshold cosmetics, is the metric on which such interventions should be designed and judged.